Ogbonna Arinze

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SUMMARY

Senior Frontend Developer with 8+ years of experience building high-impact, AI-powered web applications using React, TypeScript, and Next.js at scale. Proven expertise in designing and integrating generative AI capabilities into production products, including LLM-powered interfaces, agentic workflows, streaming responses, and real-time conversational UIs using the Vercel AI SDK, LangChain, and RAG architectures. Passionate about shaping intelligent user experiences that combine frontend excellence with cutting-edge AI, while establishing platform patterns and best practices that enable AI adoption across engineering teams.

CORE COMPETENCIES

- **React & TypeScript:** React 18/19, hooks, memoization, concurrent rendering, advanced TypeScript patterns, and 5+ years of production-grade React development for complex, data-intensive applications.
- **AI-Powered Frontend Development:** Vercel AI SDK, LangChain, Mastra, Generative UI patterns, LLM API integrations (OpenAI, Anthropic), agent frameworks, prompt engineering, and building conversational and chat-based interfaces.
- **RAG & Real-Time AI Patterns:** Retrieval-Augmented Generation strategies, vector search (Chroma, Pinecone, Weaviate), embeddings, semantic retrieval, WebSocket streaming, and handling asynchronous, long-running AI operations in the UI.
- **State Management & Data Fetching:** Redux Toolkit, React Query, Zustand, Jotai, GraphQL, REST API integration, and efficient cache invalidation for both client state and server-side data.
- **Performance & Frontend Architecture:** Code splitting, lazy loading, virtualization, bundle analysis, micro-frontends, monorepos, SSR/SSG strategies, and Core Web Vitals optimization.
- **Styling & Component Systems:** Tailwind CSS, CSS Modules, Radix UI, custom design systems, Storybook, accessibility (WCAG/ARIA), and Figma-to-code workflows.
- **Testing & Quality:** Jest, Cypress, Playwright, end-to-end and integration testing, CI-driven QA with Allure Reports, and strong commitment to clean, maintainable, well-tested code.
- **Cloud, CI/CD & Observability:** AWS (CloudFront/S3), Docker, Vercel, GitHub Actions, CI/CD pipeline automation, Sentry, Datadog RUM, and LogRocket for frontend monitoring.
- **Leadership & Product Thinking:** Technical direction, architecture reviews, product ownership, cross-functional collaboration with PMs and designers, mentorship, code reviews, and Agile/Scrum methodologies.

WORK EXPERIENCE

Roche Genentech San Francisco, CA (Remote)

Frontend Software Engineer (React/TypeScript) *May 2023 - Present*

- Architected and delivered **AI-powered frontend interfaces** by integrating LLM capabilities into a **React** and **TypeScript** clinical data platform, leveraging the **Vercel AI SDK** and **LangChain** to build intelligent features including natural language data querying, automated report generation, and context-aware recommendations for clinical researchers.
- Designed **streaming UI components** for real-time LLM responses using **WebSockets** and server-sent events, implementing token-by-token rendering, loading states, and graceful error recovery to deliver a smooth, responsive conversational experience within the product.
- Built **agentic workflow interfaces** that handle long-running, multi-step AI operations, designing scalable React patterns for asynchronous agent interactions, progress indicators, and real-time status updates that kept users informed throughout complex AI processes.
- Implemented **RAG-based retrieval interfaces** by connecting frontend components to vector search backends (embeddings, semantic retrieval), enabling users to query large clinical datasets using natural language and receive contextually relevant, AI-generated insights.
- Architected a **reusable, extensible design system** in **React** and **Radix UI** with **TypeScript**, establishing a scalable component library that improved UI consistency, accessibility, and developer velocity across multiple product teams.

- Optimized **real-time UI performance** in a **React** and **TypeScript** SPA handling large datasets by implementing **virtualization, lazy loading, code splitting**, and dynamic imports, reducing initial load times by over 40% and enhancing responsiveness through memoization and efficient CDN asset delivery.
- Scaled **state management** in a large **Next.js** application by integrating **Redux Toolkit** for complex client state and **React Query** for server-side cache, eliminating redundant re-renders and establishing consistent, maintainable data flow patterns that accelerated developer onboarding.
- Contributed to **product direction and roadmap** for AI-powered features, collaborating with Product Managers and designers to define what to build and why, translating technical AI capabilities into user-facing features that delivered measurable value to clinical teams.
- Engineered **JWT-based authentication** and **role-based access control** in a **Next.js** frontend with **TypeScript** and **React Query**, implementing secure HTTP-only cookie token handling and permission-driven UI rendering in collaboration with backend teams.
- Increased release reliability by integrating **Playwright end-to-end tests** and **Jest unit tests** into the **CI/CD pipeline**, automating test reporting with **Allure Reports** hosted on GitHub Pages, and configuring real-time Slack notifications for build quality visibility on every push.
- Modernized a **legacy Vue.js web application** into a **React-based Chrome extension**, refactoring legacy code while preserving business logic, optimizing workflows for clinical site integrations, and collaborating closely with design and backend teams to enhance adoption.

DESCASIO LTD Lagos, Nigeria
Staff Frontend Engineer *May 2021 - Apr 2022*

- Owned the core **frontend architecture** for a data-heavy **React** dashboard serving enterprise clients, implementing **virtualization** and efficient **Redux** state management to handle tens of thousands of rows with smooth scrolling and responsive interactions.
- Optimized frontend performance by profiling and resolving render bottlenecks using **React Profiler** and **Lighthouse**, reducing bundle size by **30%-40%** through **webpack-bundle-analyzer** and **rollup-plugin-visualizer**, achieving a **35% faster load time** across devices.
- Built **real-time data update capabilities** using **WebSockets** and efficient DOM reconciliation strategies, ensuring the dashboard reflected live data changes without full page reloads or unnecessary re-renders.
- Designed and implemented an **extensible component architecture** with clearly defined interfaces and composition patterns, enabling teams to add new dashboard widgets and data views without modifying the core framework.
- Developed **comprehensive testing strategies** including unit tests with **Jest**, integration tests, and end-to-end tests with **Cypress**, achieving high test coverage that significantly reduced production regressions.
- Built custom **Webpack and Rollup** build configurations optimized for the dashboard's module structure, implementing tree shaking, scope hoisting, and chunk splitting strategies that minimized bundle sizes for production deployments.
- Established engineering standards for **code quality, testing**, and deployment across the team using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing lint errors by 90% and accelerating **CI/CD** cycle times.
- Implemented **algorithmic optimizations** for client-side data filtering, sorting, and search across large datasets, applying efficient data structures to maintain sub-second response times in the UI layer.
- Collaborated with **cross-functional teams** including backend engineers, product managers, and designers to define feature requirements, establish API contracts, and deliver a polished, accessible user experience for enterprise customers.
- Mentored junior engineers and guided **frontend best practices**, conducting code reviews focused on **React** performance patterns, **TypeScript** type safety, and **testing discipline**, fostering a culture of high-quality, maintainable code across the team.

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*